

Ethan Ko

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EDUCATION

University of Michigan

May 2028

Computer Engineering BSE | GPA: 3.5

Ann Arbor, MI

- **Relevant Coursework:** Computer Architecture & Organization, Digital Logic Design, Digital Integrated Circuits, Embedded Systems, Computer Networks, Signals & Systems, Data Structures & Algorithms, Electronic Circuits
- **Activities:** Michigan Climbing Team, Michigan Pops Orchestra

EXPERIENCE

FPGA Engineering Intern

Sept. 2026 - Present

RTX - Collins Aerospace

Remote

- Will be continuing internship remotely; transitioning teams to work on the Traffic Management Computer Center.

FPGA SEPP Intern

May 2026 - Aug. 2026

RTX - Collins Aerospace

Melbourne, FL

- Engineered a Hardware-In-the-Loop testing framework for a Xilinx Artix-7 FPGA (AC701) to verify a custom Fast Fourier Transform module on the GLU-2100 GNSS Card over Ethernet.
- Reduced test execution time from 6 hours to 1 seconds over the baseline UART prototype (99% reduction).
- Built host-side Python scripts to stream stimulus data over Ethernet via TCP and analyze the returning data.
- Integrated Vivado AXI Ethernet and MIG IPs to offload network transfers, streaming data directly into DDR3.
- Developed C firmware in Vitis, utilizing FreeRTOS and the lwIP networking stack on a MicroBlaze soft processor, to parse incoming IP packets and manage DMA transactions.
- Utilized Vivado ILA and simulator to debug endianness errors between AXI-Stream and Ethernet.
- Delivered a technical presentation on results and tradeoffs to executive leadership at the corporate headquarters.

Computer Organization (EECS 370) Class Grader

Jan. 2026 - May 2026

University of Michigan - College of Engineering

Ann Arbor, MI

- Graded homework and labs for 500+ students, ensuring fair and consistent evaluation of assignments.
- Evaluated computer organization topics, including pipelining, data & control hazards, caches, and virtual memory.

PROJECTS

Superscalar RISC Processor Design

Mar. 2026 - May 2026

- Led a 5-person team to design a 2-way superscalar processor for a RISC-style ISA using SystemVerilog.
- Implemented a micro-op queue with out-of-order issue capabilities to bypass stalls caused by high-latency ops.
- Architected a split backend featuring issue, execute, and writeback stages to maximize functional unit utilization.
- Integrated G-share branch prediction alongside a 2-bank Branch Target Buffer (BTB) to mitigate control hazards.
- Developed a direct-mapped instruction cache to minimize fetch latency by exploiting spatial and temporal locality.
- Synthesized RTL onto the TSMC 250 nm PDK with Synopsys Design Compiler, achieving a max freq of 167 MHz.

Smart Doorbell Lock System

Mar. 2026 - May 2026

- Engineered a smart doorbell lock system within a 5-person group on an STM32 Nucleo, utilizing FreeRTOS.
- Integrated a keypad that drives a piezo deadbolt and a PIR motion sensor that triggers an Arducam camera.
- Stored captured images directly to an external SD card via SPI, allowing for expandable storage.
- Implemented audio playback through a speaker and amplifier using DMA to minimize CPU overhead.
- Configured a digital display via I2C for system feedback and programmed a PWM-driven buzzer for the doorbell.

TECHNICAL SKILLS

- **Languages:** VHDL, Verilog/SystemVerilog, C/C++, Arm Assembly, Python, MATLAB, Bash
- **Tools:** Vivado, Vitis, Synopsys DC, Virtuosio, STM32 IDE, Linux, Git, SVN, FreeRTOS
- **Protocols:** Ethernet, AXI, UART, I2C, SPI, UDP, TCP
- **Lab Equipment:** Oscilloscope, Multimeter, Function Generator, Power Supply, Soldering Iron